# *Project Proposal & Technical Specification (BSc Data Science Level)*

## *PROJECT TITLE: Deconstructing Greatness (The Big Three GOAT Engine)*

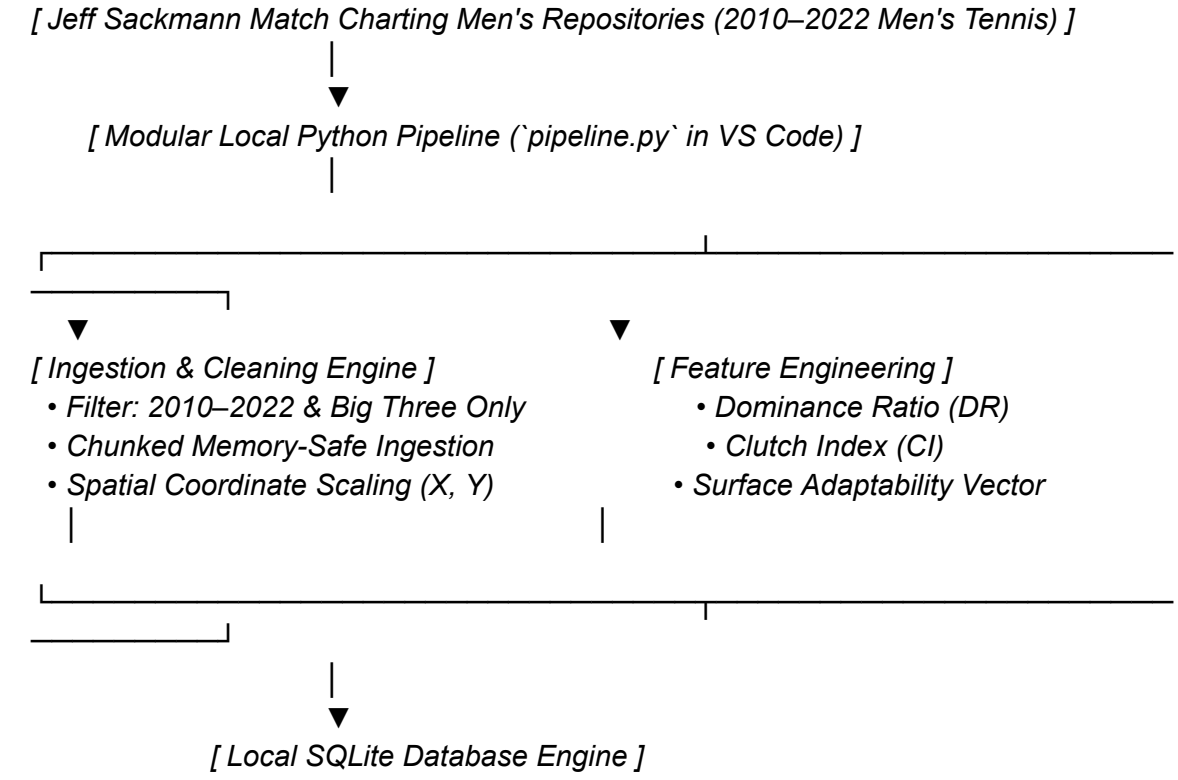
***Subtitle:*** *An End-to-End Analytics Pipeline, Spatial Feature Engineering Framework, Machine Learning Engine, and 4-Page Interactive Tableau Public Dashboard.*

## *PROJECT OVERVIEW & ARCHITECTURE*

*The objective of this project is to quantitatively address the "Greatest of All Time" (GOAT) debate among* ***Roger Federer****,* ***Rafael Nadal****, and* ***Novak Djokovic****.*

*Instead of relying on subjective commentary, this project processes* ***2010–2022 ATP match data*** *and point-by-point shot trajectory logs using a modular Python and SQL backend built entirely in VS Code. It applies Principal Component Analysis (PCA) for behavioral playstyle segmentation, trains an XGBoost classifier with Stratified 5-Fold Cross-Validation, applies 1,000-iteration bootstrapping for statistical confidence intervals, and exports normalized relational outputs to a* ***4-page interactive Tableau Public dashboard****.*

### *System Architecture Workflow*



- *(`goat_master.db`)*
- *|*
- *▼*
- *[ Machine Learning & Analytics Engine ]*
- *• Unsupervised PCA (Dimensionality Reduction: Firepower vs. Endurance)*
- *• Supervised XGBoost Match Hold Predictor (Hyperparameter Grid Search)*
- *• Stratified 5-Fold Cross-Validation (ROC-AUC Optimization)*
- *• 1,000-Iteration Non-Parametric Bootstrapping (95% CIs)*
- *|*
- *▼*
- *[ Relational CSV & SQLite Exports ]*
- *|*
- *▼*
- *[ Tableau Public Masterpiece Dashboard ]*
- *(Page 1: Overview | Page 2: Federer | Page 3: Nadal | Page 4: Djokovic)*

# *CHECKPOINT I: DATA INGESTION & LOCAL BACKEND INFRASTRUCTURE*

## *mini-checkpoint (i): Targeted Scope, Data Sources & Extraction Methodologies*

- ***a. Target Scope & Cohort:*** *Men's Professional Tennis (2010–2022) specifically filtering matches involving* ***Roger Federer****,* ***Rafael Nadal****, or* ***Novak Djokovic****. This 13-year era represents the golden peak of their rivalries and offers the highest density of crowd-sourced match charting logs.*
- ***b. Source Repository:*** *Jeff Sackmann's tennis_MatchChartingProject.*
  - *charting-m-matches.csv (Men's match metadata, dates, players, tournaments, surfaces)*
  - *charting-m-points.csv (Point-by-point serve, rally, and point-outcome logs)*
  - *charting-m-shots.csv (Shot direction, trajectory vectors, and landing zones)*
- ***c. Memory-Optimized Local Ingestion Method:*** *Python-based streaming extraction via pandas using chunked reads (chunksize=50000) and sqlite3 in src/ingestion.py. Non-Big Three and pre-2010 / post-2022 rows are filtered out before loading into local SQLite to maintain a lightweight (~30MB) database footprint.*

## *mini-checkpoint (ii): Relational Database Architecture & Validated SQLite Schema*

- ***a. Target Database:*** *data/goat_master.db running a localized SQLite relational engine.*
- ***b. Table Schema 1 (big_three_matches_2010_2022):***
- *SQL*
- *CREATE TABLE big_three_matches_2010_2022 (*

- *match_id TEXT PRIMARY KEY,*
- *match_year INTEGER,*
- *player1 TEXT,*
- *player2 TEXT,*
- *surface TEXT,*
- *tourney_name TEXT,*
- *winner TEXT*
- *);*
- 
- 
- ***c. Table Schema 2 (big_three_spatial_shots):***
- *SQL*
- *CREATE TABLE big_three_spatial_shots (*
- *shot_id INTEGER PRIMARY KEY AUTOINCREMENT,*
- *match_id TEXT,*
- *player_name TEXT,*
- *opponent_name TEXT,*
- *surface TEXT,*
- *shot_type TEXT,*
- *court_zone TEXT,*
- *shot_x REAL,*
- *shot_y REAL,*
- *is_synthetic_jitter INTEGER,*
- *is_winner INTEGER,*
- *is_unforced_error INTEGER,*
- *FOREIGN KEY (match_id) REFERENCES big_three_matches_2010_2022(match_id)*
- *);*
- 
- 
- ***d. Table Schema 3 (model_metrics_ci):***
- *SQL*
- *CREATE TABLE model_metrics_ci (*
- *execution_date TEXT,*
- *model_version TEXT,*
- *best_cv_roc_auc REAL,*
- *ci_95_lower REAL,*
- *ci_95_upper REAL*
- *);*
- 
- 

# *CHECKPOINT II: DATA CLEANING, SANITIZATION & DOMAIN STANDARDIZATION*

### *mini-checkpoint (i): Null Handling, Schema Uniformity & Type Casting*

- ***a. Missing Value Imputation:*** *Missing serve/return metrics imputed using player-surface rolling seasonal medians for the 2010–2022 timeframe rather than global means.*
- ***b. Target Variable Normalization:*** *Binary classification targets (is_hold, is_winner) cast strictly to integer types (0, 1).*
- ***c. Filtering Target Cohort:*** *Enforces strict subset filtering isolating matches where player1 or player2 belongs to ['Roger Federer', 'Rafael Nadal', 'Novak Djokovic'] within 2010 <= match_year <= 2022.*

### *mini-checkpoint (ii): Spatial Coordinate Normalization & Jittering Separation*

- ***a. Authentic Coordinate Grid:*** *Standard singles tennis court dimensions ($78ft \times 27ft$) mapped onto a Cartesian coordinate space where the net rests at $X = 0$, the server baseline rests at $X = -39$, the receiver baseline rests at $X = 39$, and sidelines span $Y \in [-13.5, 13.5]$.*
- ***b. Separation of Authentic vs. Visual Coordinates:*** *To maintain data integrity, authentic discrete zone codes from Match Charting (e.g., T-Line, Wide, Body) are preserved in shot_x and shot_y. Synthetic Gaussian jittering ($\sigma = 0.5ft$) is generated strictly for continuous density rendering in Tableau and flagged explicitly via is_synthetic_jitter = 1.*

### *mini-checkpoint (iii): Surface & Format Normalization*

- ***a. Surface Categories:*** *Standardized into three primary categories: Hard, Clay, and Grass.*
- ***b. Best-of-3 vs. Best-of-5 Scaling:*** *Normalizing endurance metrics across Master 1000s (Best-of-3) and Grand Slams (Best-of-5) using per-set rate metrics instead of raw match totals.*

## *CHECKPOINT III: DOMAIN-SPECIFIC FEATURE ENGINEERING & STATISTICAL METRICS*

### *mini-checkpoint (i): Advanced Custom Analytics & Protected Formulas*

- ***a. Protected Dominance Ratio (DR):*** *Ratio of return points won percentage to service points lost percentage, protected against division-by-zero errors via epsilon floor ($\epsilon = 1e-5$):*
  *$$\text{DR} = \frac{\% \text{ Return Points Won}}{\max(100\% - \% \text{ Service Points Won}, \epsilon)}$$*
  *Concept: Measures total match control. A $DR > 1.0$ indicates superior court control; a $DR > 1.4$ represents dominant performance.*

- ***b. Clutch Index (CI / Pressure Resistance):*** *Measures performance under high-stress conditions (where* ***BP*** *= Break Points):*
    - $CI = \left(\frac{BPSaved}{BPFaced} \times 0.5\right) + \left(\frac{TiebreaksWon}{TiebreaksPlayed} \times 0.3\right) + \left(\frac{DecidingSetsWon}{DecidingSetsPlayed} \times 0.2\right)$
- *Concept: Quantifies psychological resilience during critical leverage points.*
- ***c. Surface Adaptability Index (SAI):*** *Standard deviation of a player's win rate across Hard, Clay, and Grass courts (2010–2022):*
    - $SAI = 1 - \sqrt{\frac{1}{3}\sum_{surface}(WinRate_{surface} - WinRate_{overall})^2}$
- *Concept: Values closer to* $1.0$ *indicate an all-court playing style (high versatility), while lower values indicate surface specialization.*
- ***d. Tactical Innovation & Adaptability Score (%):*** *A dynamic composite metric (0–100%) evaluating tactical adaptation across three sub-eras (2010–2014, 2015–2018, 2019–2022):*
- $InnovationScore = (w_1 \cdot \Delta NetFreq) + (w_2 \cdot \Delta ServeSpeedVar) + (w_3 \cdot \Delta DropShotFreq) + (w_4 \cdot Rac$

# *CHECKPOINT IV: EXPLORATORY DATA ANALYSIS (EDA) & UNSUPERVISED ML (PCA)*

## *mini-checkpoint (i): Exploratory Data Analysis & Feature Standardization*

- ***a. Feature Standardization (StandardScaler):*** *Prior to PCA, all 15 continuous features are standardized to zero mean and unit variance (*$\mu = 0, \sigma = 1$*) to prevent high-magnitude features (e.g., total service points) from artificially dominating variance.*
- ***b. 2010–2022 Peak Era Dominance Profiles:*** *Time-series analysis evaluating peak rolling 3-year win percentages and Grand Slam conversion rates to isolate each player's absolute peak within the decade.*
- ***c. Non-Technical Translation Mapping:***
    - *1st Serve Win % →* ***First-Try Conversion Rate***
    - *Break Points Saved (BP Saved) →* ***Crisis Recovery Under Pressure***
    - *Surface Adaptability (SAI) →* ***Environmental Versatility***

## *mini-checkpoint (ii): Unsupervised Learning — Principal Component Analysis (PCA)*

- ***a. Dimensionality Reduction Purpose:*** *Compresses 15 continuous match features down to two principal components.*
- ***b. Component Interpretability:***
    - ***PC1 (Service Firepower & Attack):*** *Explains variance driven by free points on serve, net approaches, and short rally duration (High = Federer).*

    - ***PC2 (Defensive Endurance & Rally Resilience):*** *Explains variance driven by long rally tolerance, return points won, and baseline groundstroke consistency (High = Nadal & Djokovic).*
- ***c. Player Archetype Clustering:*** *Mapping all ATP players onto the 2D Principal Component space to demonstrate that the Big Three occupy distinct, non-overlapping performance personas.*

# *CHECKPOINT V: SUPERVISED MACHINE LEARNING, LEAKAGE PREVENTION & BOOTSTRAPPING*

## *mini-checkpoint (i): Predictive Model Framing & Target Leakage Prevention*

- ***a. Predictive Boundary Enforcement:*** *To eliminate target leakage, models predicting game $G_t$ only utilize historical match logs strictly prior to the match date $(t-1)$ and game-start state vectors. Match-level aggregates from the target match are excluded from input vectors.*
- ***b. Algorithm Selection: XGBoost Classifier (xgb.XGBClassifier)*** *selected for its handling of non-linear feature interactions and high gradient boosting efficiency.*

## *mini-checkpoint (ii): Hyperparameter Testing & Cross-Validation Framework*

- ***a. Validation Strategy: Stratified 5-Fold Cross-Validation (StratifiedKFold)*** *to handle class balance across surfaces and prevent target leakage.*
- ***b. Hyperparameter Search Space (GridSearchCV):***
    - *max_depth: [3, 5, 7]*
    - *learning_rate: [0.01, 0.05, 0.1]*
    - *n_estimators: [50, 100, 200]*
    - *subsample: [0.8, 1.0]*
    - *colsample_bytree: [0.8, 1.0]*
- ***c. Optimization Metric:*** *Explicitly scored via scoring='roc_auc'.*

## *mini-checkpoint (iii): Statistical Rigor — Non-Parametric Bootstrapping*

- ***a. Bootstrapping Methodology:*** *Executing **1,000 empirical resamples** with replacement from the test set predictions.*
- ***b. Confidence Interval Calculation:*** *Computing the 2.5th and 97.5th percentiles of predicted win probabilities to establish formal **95% Confidence Intervals** around head-to-head predictions.*
- ***c. Integration Target:*** *Exporting lower and upper CI bounds directly into SQLite/CSV so error bars render natively on Tableau Public.*

# *CHECKPOINT VI: EXPORT ENGINE & TABLEAU PUBLIC MASTERPIECE LAYOUT*

## *mini-checkpoint (i): Export & Simulator Offline Logic Architecture*

- ***a. Multi-Table CSV Generation:*** *Python pipeline outputs three cleaned CSV extracts:*
    1. *data/tableau_goat_overview.csv (Aggregated KPIs, 2010–2022 career stats, brand deals, philanthropy).*
    2. *data/tableau_goat_spatial.csv (Authentic vs. synthetic jittered coordinates $X, Y$).*
    3. *data/tableau_goat_predictions.csv (XGBoost probabilities, response surface matrix, 95% CIs).*
- ***b. Tableau "GOAT Simulator" Logic Disconnect Resolution:*** *Tableau Public cannot execute real-time Python/XGBoost inference. The simulator operates using a* ***pre-calculated response surface grid*** *exported by Python across all parameter permutations, mapped dynamically using Tableau Parameter weighted arithmetic.*

## *mini-checkpoint (ii): Design System & Aesthetics*

- ***a. Visual Theme:*** *"Swiss Precision Dark Mode" palette.*
    - *Canvas Background: Navy/Slate (#0D1117)*
    - *Container Background: Glassmorphism Dark Blue (#161B22)*
    - *Text Headers: Pure White (#FFFFFF) / Muted Gray (#8B949E)*
    - *Accent Colors: Gold (#FFD700), Federer Crimson (#E63946), Nadal Clay Orange (#E76F51), Djokovic Neon Green (#2A9D8F).*
- ***b. Canvas Dimensions:*** *Fixed 1920x1080 resolution per page with consistent 10px padding.*

## *mini-checkpoint (iii): Detailed 4-Page Tableau Dashboard Specification*

### *PAGE 1: OVERVIEW & THE GOAT SIMULATOR*

- ***Section A (Hero Header & Profiles):*** *High-res player portraits, bio metadata (height, weight, dominant hand, forehand/backhand style).*
- ***Section B (2010–2022 Master KPI Table):*** *Overall win-loss, weeks at #1, prize money earnings, total titles split by surface (Hard, Clay, Grass).*
- ***Section C (PCA Archetype Map):*** *Scatter plot mapping PC1 (Firepower) vs PC2 (Endurance) with player position clusters.*
- ***Section D (Interactive GOAT Simulator):*** *Parameter controls allowing users to select 1, 2, or 3 players and adjust weighting sliders. Outputs a dynamic* ***Dominance Score*** *with* ***95% Confidence Intervals*** *rendered via offline response surface mapping.*

### *PAGE 2: ROGER FEDERER PROFILE ("The Maestro")*

- ***Section A (Header KPIs):*** *Name, Picture, 310 Weeks at #1, $130.6M Earnings, Peak Dominance Sub-Era:* ***2010–2012 & 2017****.*
- ***Section B (Court Shot Placement Heatmap):*** *Spatial density map showing shot distribution on a custom court layout, highlighting high net-approach frequency.*
- ***Section C (Serve Placement Heatmap):*** *Landing zone distribution for 1st and 2nd serves (T-line vs. Wide Slice).*

- ***Section D (Tactical Analysis):*** *Innovation Level Score (**94%** - SABR attack, racquet head change), Playstyle Breakdown (Aggressive All-Court), Surface Win % (Grass/Hard emphasis).*
- ***Section E (Off-Court Legacy):*** *Brand Partnerships (Uniqlo, Rolex, On Running) & Philanthropy (**Roger Federer Foundation**).*

***PAGE 3: RAFAEL NADAL PROFILE ("The King of Clay")***

- ***Section A (Header KPIs):*** *Name, Picture, 209 Weeks at #1, $134.9M Earnings, Peak Dominance Sub-Era: **2010–2013 & 2017–2020**.*
- ***Section B (Court Shot Placement Heatmap):*** *Spatial density map highlighting heavy baseline positioning (6–10 ft behind baseline) and topspin cross-court angles.*
- ***Section C (Serve Placement Heatmap):*** *Serve placement map showcasing Ad-court wide slice setups.*
- ***Section D (Tactical Analysis):*** *Innovation Level Score (**91%** - Transition to aggressive hard-court play, serve mechanical adjustments), Playstyle Breakdown (Heavy Topspin Counter-Puncher), Surface Win % (Clay emphasis - 90.5%).*
- ***Section E (Off-Court Legacy):*** *Brand Partnerships (Nike, Babolat, Richard Mille) & Philanthropy (**Rafa Nadal Foundation**).*

***PAGE 4: NOVAK DJOKOVIC PROFILE ("The Machine")***

- ***Section A (Header KPIs):*** *Name, Picture, 428 Weeks at #1, $185.0M Earnings, Peak Dominance Sub-Era: **2011–2016 & 2018–2022**.*
- ***Section B (Court Shot Placement Heatmap):*** *Spatial density map demonstrating baseline depth targeting (within 1 ft of baseline) and wide sliding court coverage.*
- ***Section C (Serve Placement Heatmap):*** *Multi-directional serve placement map (unpredictable 4-corner distribution).*
- ***Section D (Tactical Analysis):*** *Innovation Level Score (**97%** - Sliding mechanics on hard courts, serve optimization under Ivanišević), Playstyle Breakdown (Aggressive Counter-Puncher / Return Specialist), Surface Win % (All-court balance).*
- ***Section E (Off-Court Legacy):*** *Brand Partnerships (Lacoste, Asics, Head, Hublot) & Philanthropy (**Novak Djokovic Foundation**).*

# *CHECKPOINT VII: MODULAR CODE EXECUTION SPECIFICATIONS & PRESENTATION*

## *mini-checkpoint (i): Python Code Execution Specifications*

| ***Module Path*** | ***Primary Class / Functions*** | ***Key Inputs*** | ***Expected Outputs / Artifacts*** |
|---|---|---|---|

| | | | |
|---|---|---|---|
| *src/ingestion.py* | *ingest_and_filter_match_charting_data()* | *Jeff Sackmann GitHub CSV URLs* | *Chunked, filtered DataFrames & SQLite tables* |
| *src/cleaning.py* | *clean_matches(), normalize_coordinates()* | *Raw SQLite tables* | *Imputed & parsed SQLite tables* |
| *src/features.py* | *compute_dominance_ratio(), compute_clutch_index()* | *Cleaned Match Data* | *engineered_features DataFrame* |
| *src/pca_clustering.py* | *run_pca_pipeline()* | *Standardized 15 continuous features* | *PC1/PC2 coordinates & variance ratios* |
| *src/ml_model.py* | *train_xgboost(), run_bootstrapping()* | *Engine Features & Historical Labels ($t-1$)* | *Model metrics, ROC-AUC score, 95% CIs* |

***mini-checkpoint (ii): GitHub Repository Layout***

*Plaintext*

- *tennis-goat-analytics-engine/*
- *├── data/*
- *│ ├── goat_master.db*
- *│ ├── tableau_goat_overview.csv*
- *│ ├── tableau_goat_spatial.csv*
- *│ └── tableau_goat_predictions.csv*
- *├── src/*
- *│ ├── __init__.py*
- *│ ├── ingestion.py*

- *│   ├── cleaning.py*
- *│   ├── features.py*
- *│   ├── pca_clustering.py*
- *│   └── ml_model.py*
- *├── notebooks/*
- *│   └── exploratory_analysis.ipynb*
- *├── pipeline.py*
- *├── requirements.txt*
- *└── README.md*

## *mini-checkpoint (iii): Executive Summary & LinkedIn Pitch*

- ***a. Non-Technical Story:*** *Framing the project as an automated analytics framework that translates complex athletic performance into relatable business concepts like environmental adaptability, risk management under pressure, and long-term asset optimization.*
- ***b. Executive Resume Bullet:***
  *"Engineered an end-to-end Python & SQL sports analytics engine in VS Code processing 2010–2022 Match Charting logs; applied PCA for player segmentation, trained an XGBoost win-probability model with Stratified 5-Fold Cross-Validation and 1,000-iteration bootstrapping, delivering a 4-page interactive Tableau Public dashboard."*
-